BASE CHANGE FOR THE PUSHFORWARD ALONG AN AFFINE MORPHISM, AND QUASI-COHERENCE OF THAT PUSHFORWARD

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file spaces-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the pinned chapter file opens with \input{preamble} and carries no preamble text of its own, so the project's single Stacks-wide preamble staged at sources/stacks-spaces-cohomology/preamble.tex is excerpted verbatim as three spans, lines 8-12 (verbatim package and the project's own reference/slogan/history display environments), lines 14-18 (Xy-pic setup) and lines 154-217 (theorem declarations and mathematical macros). The wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler. No macro was invented or redefined.

Editorial note. Same-chapter pointers are located at their \label lines inside the pinned chapter file; cross-chapter pointers are located in the project's own tag list staged in the snapshot (sources/stacks-chow/tags.txt), at the verbatim lines carrying their canonical tags.

Allowed standard prerequisites (editorial).

- The base change diagram for higher direct images. Same chapter spaces-cohomology.tex, equation equation-base-change-diagram (cited at statement line 1906). It fixes the cartesian diagram to which the asserted equality $g^*f_*\mathcal{F} = f'_*(g')^*\mathcal{F}$ refers.
- The general representable case of the higher direct images. Same chapter spaces-cohomology.tex, the discussion surrounding equation-representable-higher-direct-image (invoked at proof line 1915). It reduces the statement for algebraic spaces to the corresponding statement for an affine morphism of schemes, because an affine morphism is representable.
- The scheme case of base change for an affine morphism. Cohomology of Schemes, label coherent-lemma-affine-base-change (invoked at proof line 1917). It supplies the equality and the quasi-coherence in the scheme case, which is then pulled back to algebraic spaces.

Source reference. Original lemma 02KG: sources/stacks-chow/tags.txt, source lines 3371–3371. Navigation pointer for \ref{coherent-lemma-affine-base-change} used inside the counted span (line 1918); the argument that consumes it is retained in the span.

Source reference. Original equation 073J: sources/stacks-spaces-cohomology/pinned-original-source.tex, source lines 1885–1885. Navigation pointer for \ref{equation-base-change-diagram} used inside the counted span (line 1906); the argument that consumes it is retained in the span.

Date: stacks-project:spaces-cohomology.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original equation 071Z: sources/stacks-spaces-cohomology/pinned-original-source.tex, source lines 318–318. Navigation pointer for \ref{equation-representable-higher-direct-image} used inside the counted span (line 1915); the argument that consumes it is retained in the span.

Lemma 0.1. *Let S be a scheme. Let $f : X \to Y$ be an affine morphism of algebraic spaces over S. Let $\mathcal{F}$ be a quasi-coherent $\mathcal{O}_X$-module. In this case $f_*\mathcal{F} \cong Rf_*\mathcal{F}$ is a quasi-coherent sheaf, and for every diagram (073J) we have*

$$g^* f_*\mathcal{F} = f'_*(g')^*\mathcal{F}.$$

Proof. By the discussion surrounding (071Z) this reduces to the case of an affine morphism of schemes which is treated in Cohomology of Schemes, Lemma 02KG. □